

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java, C/C++

Backend & APIs: FastAPI, Flask, Node.js/Express, REST/JSON, WebSockets, LLM integration, input validation, schema design

Frontend: React, HTML/CSS/JavaScript, client-side state management, responsive UI

Data & Quality: SQLite, SQL Server, pandas, NumPy, pytest, Jest, GitHub Actions, Docker, structured logging, Git, Postman

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning for Python telemetry tooling, controller validation, CAN diagnostics, and shared testing workflows
- Standardize project requirements, test evidence, and technical reviews across dashboard, simulation, and vehicle software projects

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated flight-control software in C on PX4, using structured telemetry logs to reproduce issues and evaluate changes
- Integrated visual-inertial odometry and iterated configuration through repeatable tests and documented results

Software Development Intern

Sep 2024 - Oct 2024

Zip Intelligence | Tracy, CA

- Implemented frontend validation and clearer error states, then paired with engineers to harden REST endpoints for status codes and edge cases
- Added SQL Server update scripts and structured logs to speed reproduction and issue triage
- Took ownership of a small feature from ticket through verified release, shipping via pull requests with setup and change documentation

PROJECT EXPERIENCE

Tickr — AI-Coached Paper-Trading Game

2025

TypeScript/JavaScript, React, LLM integration, flat-file datastore, responsive UI

- Built registration, sign-in, game, and history flows for a scenario-driven trading game with an AI coach
- Implemented quest and prompt logic, client-side state management, and a mobile-responsive interface

SR-Wireless-CAN — Wireless Vehicle Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, FastAPI, WebSockets, React, SQLite, Docker Compose

- Led backend development for a wireless vehicle telemetry platform, building a FastAPI and WebSocket service that streams live data to a React dashboard with SQLite persistence
- Implemented structured upload and merge of signal definitions with automatic conflict resolution, and a bounded search window keeping large catalogs responsive
- Reverse-engineered an undocumented device flashing protocol from bus trace captures and reimplemented it in Python, enabling wireless firmware updates

Scent Showdown

2025

Flask, HTML/CSS/JavaScript, SQLite, REST/JSON

- Built a matchup and leaderboard application with Flask routes, templates, a compact SQLite schema, input validation, and an admin view
- Added JSON endpoints, seed and export scripts, deployment setup, and a maintainer-focused README